

HOW WE GROW BIG BRAIN

Counterfactual Generative Morphogenesis (Explained for Cavemen)



14 Apes. One Cave. One Job Each.

USE OLD ROCK **(Fixed Blueprints)**



- Rock is already made.
- Rock is usually wrong shape for new problem.
- One ape tries to find rock, hit rock, and put rock in brain. Ape gets confused.

GROW NEW BRAIN PART **(Generative Morphogenesis)**



- Look at exact problem first.
- Make fresh rock that fits hole perfectly.
- Many apes. Each ape does one tiny job perfectly.

THE GOLDEN TRIBE RULE

ONE APE, ONE JOB.



- Scout only looks.
Scout never makes
rock.

- Caller only points.
Caller never
makes rock.

- Maker only makes
rock. Maker never
sees the target.

If one ape tries to look, think, and hit all at once, the Big Brain breaks.



THE BIG RULES (Leyline)

How apes talk. What is allowed.



PROVING GROUNDS (Tolaria)

Where we test rocks and train the Big Brain.



MEMORY WALL (Sarpadia)

We draw every good rock, every bad rock, and every dead ape here. We never forget.



THE CHIEF

(Tamiyo)

DO:

- Decide how much food and time the tribe has.
- Pick which parts of Big Brain get attention.

DO NOT:

- Tell anyone what kind of rock to make.



THE CALLER

(Narset)

DO:

- Point at a hole in the Big Brain and say "We need a rock here right now!"

DO NOT:

- Tell the Maker what shape the rock should be. (Do not pre-write the answer).



THE SCOUT

(Nissa)

DO:

- Look deeply at the Big Brain.
- Yell exactly what is happening loudly to everyone.
- Tell the exact truth.

DO NOT:

- Say 'I think we should build this.'
- Do not tell anyone how to fix the problem.
- Just show the problem.



THE MAKER

(Momir)

DO:

- Look at Scout's drawing. Invent a new, raw rock that might fit.

DO NOT:

- Test the rock yourself.



THE SMOOTHER

(Elesh)

DO:

- Check if Maker's rock breaks The Big Rules. Make the edges legal.

DO NOT: Care if the rock is actually useful. Only care if it is safe.



THE SHARPENER

(Tezzeret)

DO:

- Bind the rock to a stick so it is ready to be used in the Proving Grounds.

DO NOT: Change the shape Maker and Smoother agreed on.

THE HITTER (Urabrask)



DO: Smash the new rock in the Proving Grounds. Record exactly what happens.

DO NOT: Say if the rock is good enough to keep.



**WHAT IF WE
DO NOTHING?**

THE ELDER (Augustin)



DO: Look at the smashed rock. Compare it to the "Do Nothing" timeline. Say YES or NO.

DO NOT: Smash the rock yourself. (Judges do not swing clubs).

THE HEALER (Kasmina)



DO: If Elder says YES, carefully weave the new rock into the Big Brain. Make it blend slowly so Brain does not go crazy.

DO NOT: Pick which rock is good. Only weave what is approved.



THE REAPER (Emrakul)



DO: Watch old rocks. If a rock stops working, put it to sleep. Rip it out safely so the Brain can grow elsewhere.

DO NOT: Make new rocks to replace the dead ones.



THE PAINTER

(Oona)

DO:

- Watch the entire hunt.
- Paint every single thing that happens on the wall so outsiders can see our story.

DO NOT:

- Touch the Big Brain.
- If Painter goes to sleep, the tribe must keep hunting exactly the same way.



**1. Big Brain
is hurting.**



**2. THE SCOUT
(Nissa) takes a
picture of the gap.
Sends picture to
Caller and Maker.**



**3. THE CHIEF
(Tamiyo) says
“We have enough
food to fix this.”**



**4. THE CALLER
(Narset) yells
“Maker, we need
a fix here! Hurry!”**

RULE CHECK: The Caller points, but does NOT tell the Maker what shape to make.

5. THE FORGE



-  - **MAKER** invents rock.
-  - **SMOOTHER** makes it legal.
-  - **SHARPENER** puts it on a stick.

6. PROVING GROUNDS



Hit problem with
the New Rock.

Hit problem with
'Do Nothing'
(Empty Hand).



7. THE ELDER looks at
both paths. Does New
Rock beat 'Do Nothing'?



8. THE HEALER (Kasmina) weaves the winning rock into the Big Brain. Brain gets stronger.

9. Time passes. Rock gets old and lazy.

11. Every single step is carved into the MEMORY WALL (Sarpadia). We learn from dead rocks just as much as living ones.

10. THE REAPER (Emrakul) puts the old to sleep and rips it out.

TAKING OFF THE TRAINING WHEELS

(Scaffold Withdrawal)

Maker looks at rocks the ancestors made. Copies them to learn how rock works.

1. Look at Old Rock

(Bootstrap Ancestry)

Maker takes an ancestor's rock and makes it slightly sharper or wider.

2. Change Old Rock

(Mutation)

Maker closes eyes, remembers the Big Rules, and invents a shape never seen before. It beats the Old Rock.

3. Throw Old Rock Away

(Scaffold-Free Generation)

BIG BRAIN GROWS. THE CAVE REMEMBERS.

We do not guess.
We do not use old rocks
for new problems.
We build exactly what is
needed, proven it beats



...had weate it into life
We build exactly
what is needed, prove
it beats doing nothing,
and weave it into life.

One Ape. One Job. No Exceptions.